

AMCS2034

Tutorial 3: Algorithms

- Q1. Describe **TWO (2)** types of containers that the Java Collection Framework supports.
- Q2. What does each of the collection listed below store?
- (i) Sets
 - (ii) Lists
 - (iii) Stacks
 - (iv) Queues
 - (v) PriorityQueuees
- Q3. What is an iterator?
- Q4. Briefly explain the usage of any **THREE (3)** methods belonging to the Iterable interface.
- Q5. How were design patterns obtained?
- Q6. State the **TWO (2)** principles of object-oriented design that is used for creating design patterns.
- Q7. Discuss the **THREE (3)** steps in implementing an Iterator based on the Iterator Pattern.
- Q8. Define *Strings* as used in the Java programming language.
- Q9. In the Java programming language, the *String* class is immutable.
- (i) Briefly explain the meaning of immutability in terms of the Java String class
 - (ii) Which classes would be more suitable to use if the stored string need to be modified during runtime?
- Q10. Dynamic programming is a problem solving technique similar to the greedy, and divide and conquer approaches. Describe the Dynamic programming approach to problem solving.
- Q11. Describe **ONE (1)** benefit of using Dynamic programming.
- Q12. Explain the **FOUR (4)** steps of performing Dynamic programming in solving a problem.
- Q13. Describe the **THREE (3)** classes belonging to the java.util.regex package.